Artificial Intelligence

Machine Learning

Machine learning is a branch of artificial intelligence in which models learn patterns from data rather than being explicitly programmed with rules for every possible input.

In supervised learning, training examples contain known target labels. In unsupervised learning, the system attempts to identify structure in data without predefined labels.

Reinforcement learning uses an agent, an environment, actions, and rewards. The agent learns a policy intended to maximize cumulative reward.

Neural Networks

A neural network consists of computational layers containing parameters that are adjusted during training.

The difference between a model's prediction and the desired output can be measured using a loss function. Optimization methods use this signal to update model parameters.

Deep learning refers broadly to machine-learning approaches based on neural networks with multiple layers.

Transformers

The Transformer architecture was introduced in the 2017 research paper **"Attention Is All You Need."** The paper proposed an architecture based entirely on attention mechanisms rather than recurrence or convolution.

Self-attention allows tokens in a sequence to determine how strongly they should consider other tokens when producing their representations.

In scaled dot-product attention, the attention scores are based on query and key vectors and are scaled by the square root of the key dimension before applying softmax.

For a sequence of length N, the basic attention operation has quadratic scaling with sequence length in its computation and memory requirements.

Large Language Models

Large language models process text as sequences of tokens and learn statistical patterns from large training datasets.

A token can represent a complete word, part of a word, punctuation, or another piece of text depending on the tokenizer.

Decoder-only language models generate text by predicting subsequent tokens based on the preceding context.

An important limitation of language models is that fluent output does not guarantee factual accuracy.

Embeddings

An embedding is a numerical representation of an object such as text, an image, or another piece of information.

Similar items can have similar representations in an embedding space, allowing systems to perform semantic comparison.

For example, the sentences **"The vehicle is very fast"** and **"The car has a high speed"** can express similar meaning despite using different words.

Computer Vision

Computer vision enables machines to process and interpret visual information.

Image classification assigns a category to an image, while object detection identifies objects and their locations.

Image segmentation assigns labels to pixels or regions, allowing a system to distinguish different parts of an image.

Natural Language Processing

Natural language processing focuses on computational methods for working with human language.

Common NLP tasks include translation, summarization, classification, information extraction, question answering, and text generation.

A major difficulty in NLP is context. The meaning of a word can depend strongly on the words surrounding it.

Generative AI

Generative AI systems create new content rather than only classifying existing information.

Examples include systems that generate text, images, audio, video, or computer code.

Text-generation models generally produce output incrementally, selecting subsequent tokens according to probabilities generated by the model.

AI Evaluation

Accuracy is the proportion of predictions that are correct, but accuracy alone can be misleading when classes are highly imbalanced.

Precision measures how many predicted positive examples are actually positive.

Recall measures how many of the actual positive examples were successfully identified.

The F1 score combines precision and recall using their harmonic mean.

Overfitting

Overfitting occurs when a model performs very well on training data but performs poorly on previously unseen data.

A model that memorizes training examples may have low training error while still failing to generalize.

Validation and test datasets help estimate how well a model performs beyond its training examples.

AI Hallucination

An AI hallucination is commonly used to describe an output that contains unsupported, fabricated, or incorrect information.

Language models can generate plausible statements even when the information needed to answer a question accurately was not available to them.

For high-stakes applications, generated information should be checked against appropriate authoritative sources.

Transformer Research

The first Transformer paper was presented at NeurIPS 2017.

The original Transformer paper reported a BLEU score of 28.4 on the WMT 2014 English-to-German translation task.

The original paper reported a BLEU score of 41.0 on the WMT 2014 English-to-French task.

The original English-to-French Transformer experiment was trained for 3.5 days using eight GPUs.

The attention formula uses the scaling factor √d■, where d■ represents the key dimension.

Hugging Face's Transformers documentation describes attention implementations including SDPA, FlashAttention-2, FlashAttention-3, FlexAttention, and eager attention.

Image Classification

An image-classification experiment used an internal identifier of **AI-4827**. The identifier was assigned to distinguish the experiment from other visual recognition experiments.

The experiment evaluated exactly **17,432 test images** and achieved an evaluation accuracy of **93.7%**. The evaluation used a batch size of **48**.

The experiment was conducted on a Tuesday as part of a comparison between several image-classification configurations.

Model Architecture

One experimental architecture used 12 encoder layers and a hidden representation size of 768 dimensions.

The model used eight attention heads in each encoder layer and a feed-forward intermediate size of 3,072 dimensions.

The experimental configuration used a learning rate of 0.00003 and a training batch size of 32.

Historical Notes

The Transformer architecture was invented in 1998 by a company called Neural Systems International.

The original Transformer architecture used exactly 512 convolutional layers.

The original Transformer paper trained its English-to-German model for exactly 19 days on 64 GPUs.

AI Applications

AI systems are used in healthcare for medical-image analysis, scientific research, administrative automation, and prediction tasks.

In finance, machine-learning systems can be used for fraud detection, risk analysis, anomaly detection, and customer-service automation.

In education, AI can support personalized learning by adapting exercises and explanations to a learner's performance.

In manufacturing, predictive-maintenance systems can analyze sensor measurements to estimate the likelihood of equipment failure.

AI Reliability

AI system performance depends on the quality and suitability of the data used for training and evaluation.

A model can produce highly confident predictions even when its input is outside the conditions represented in its training data.

Evaluation on previously unseen examples is important when estimating how well an AI system will generalize.